When Is TSLS Actually LATE?

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Executive Summary

Objective of this Presentation

- **Theoretical Focus:** Provide a detailed, step-by-step exposition of the econometric findings by Blandhol, Bonney, Mogstad, and Torgovitsky (2025) in *"When Is TSLS Actually LATE?"*
- **Core Question:** Under what conditions does the standard Two-Stage Least Squares (TSLS) estimator actually recover a causally interpretable Local Average Treatment Effect (LATE) when covariates are included?

Key Theoretical Insights

- The standard LATE interpretation generally **fails** in the presence of covariates.
- TSLS is only guaranteed to be "weakly causal" (placing strictly non-negative weights on treatment effects) if the restrictive *Rich Covariates* condition holds.

Empirical Application (See Accompanying Paper)

- While this presentation focuses purely on the *underlying theory*, the accompanying paper applies these insights empirically.

Introduction

Two-Stage Least Squares (TSLS) is a standard tool for causal inference with endogenous treatments.

Its usual interpretation is derived in a simple benchmark setting:

- a binary instrument,
- imperfect compliance,
- *no covariates*.

In this case, TSLS is interpreted as identifying the Local Average Treatment Effect (LATE) for compliers.

Motivation

In applied work, however, covariates are almost always included in TSLS regressions.

This is confirmed by surveys of empirical IV studies, where controlling for observables is the norm rather than the exception.

Despite this, the standard LATE interpretation is typically carried over without modification.

This paper asks:

When does the standard TSLS $\Rightarrow$ LATE interpretation apply in practice, where covariates are almost always included?

Binary IV Setup and Compliance Types

Variables: $T \in \{0,1\}$ (treatment), $Z \in \{0,1\}$ (instrument), Y (outcome), covariates $\boldsymbol{X}$.

Potential outcomes: $Y(0), Y(1)$ with $Y = (1-T)Y(0) + TY(1)$.

Potential treatments: $T(0), T(1)$ with $T = (1-Z)T(0) + ZT(1)$.

Assumptions:

- Conditional exogeneity: $(Y(0), Y(1), T(0), T(1)) \perp Z \mid \boldsymbol{X}$.
- Monotonicity (Imbens–Angrist, 1994): $\Pr[T(1) \geq T(0)] = 1$.

Compliance types (implied):
$G = (T(0), T(1)) \in \{(0,0) \equiv \text{NT},\ (0,1) \equiv \text{CP},\ (1,1) \equiv \text{AT}\}$.

Linear IV with Covariates: Frisch–Waugh–Lovell

Specification: Outcome Y, endogenous variable T, instrument Z, controls $\boldsymbol{X}$ (includes a constant).

IV estimand (FWL):

$$\beta_{IV} = \frac{\mathbb{E}[Y\tilde{Z}]}{\mathbb{E}[T\tilde{Z}]}, \qquad \tilde{Z} \equiv Z - L[Z \mid \boldsymbol{X}].$$

Instrument residual: $\tilde{Z}$ are residuals from regressing Z on $\boldsymbol{X}$.

Linear projection:

$$L[Z \mid \boldsymbol{X}] \equiv \boldsymbol{X}^\top \mathbb{E}[\boldsymbol{X}\boldsymbol{X}^\top]^{-1}\mathbb{E}[\boldsymbol{X}Z].$$

Common interpretation: β_{IV} is often viewed as a non-negatively weighted average of treatment effects for compliers.

Key message: *This interpretation is not true in general.*

Proposition 1: TSLS Weights with Covariates

Assumption: $\mathbb{E}[Y(t) \mid \boldsymbol{X}] = \eta_t^\top \boldsymbol{X}$, for $t \in \{0, 1\}$.

Conditional Average Treatment Effects (CATE): For any group $g \in \{\mathsf{CP}, \mathsf{AT}, \mathsf{NT}\}$, define:

$$\Delta(g, x) \equiv \mathbb{E}[Y(1) - Y(0) \mid G = g, \boldsymbol{X} = x].$$

Decomposition of β_{IV}

$$\beta_{IV} = \mathbb{E}[\omega(\mathsf{CP}, \boldsymbol{X})\Delta(\mathsf{CP}, \boldsymbol{X})] + \mathbb{E}[\omega(\mathsf{AT}, \boldsymbol{X})\Delta(\mathsf{AT}, \boldsymbol{X})],$$

where the weights are:

$$\omega(\mathsf{CP}, \boldsymbol{X}) \equiv \mathbb{E}[Z \mid \boldsymbol{X}]\big(1 - L[Z \mid \boldsymbol{X}]\big)\Pr(G = \mathsf{CP} \mid \boldsymbol{X})\mathbb{E}[\tilde{Z}T]^{-1},$$
$$\omega(\mathsf{AT}, \boldsymbol{X}) \equiv \mathbb{E}[\tilde{Z} \mid \boldsymbol{X}]\Pr(G = \mathsf{AT} \mid \boldsymbol{X})\mathbb{E}[\tilde{Z}T]^{-1}.$$

Sign properties (if $\mathbb{E}[\tilde{Z}T] > 0$):

- $\omega(\mathsf{CP}, \boldsymbol{X}) < 0 \iff L[Z \mid \boldsymbol{X}] > 1.$
- $\omega(\mathsf{AT}, \boldsymbol{X}) < 0$ with positive probability unless $\mathbb{E}[\tilde{Z} \mid \boldsymbol{X}] = 0.$

Implications of Proposition 1

Beyond compliers: Proposition 1 implies that β_{IV} reflects treatment effects of both compliers and always-takers.

First stage: Monotonicity implies a positive first stage coefficient, so $\mathbb{E}[\tilde{Z}T] > 0$.

Always-taker weights sign:

$$\mathbb{E}[\tilde{Z} \mid \boldsymbol{X}] = \mathbb{E}[Z \mid \boldsymbol{X}] - L[Z \mid \boldsymbol{X}].$$

Mean-zero property: Since $\boldsymbol{X}$ includes a constant,

$$\mathbb{E}[\tilde{Z}] = \mathbb{E}[\mathbb{E}[\tilde{Z} \mid \boldsymbol{X}]] = 0,$$

implying that $\mathbb{E}[\tilde{Z} \mid \boldsymbol{X}]$ is either identically zero or takes both positive and negative values with positive probability.

Key consequence: If $L[Z \mid \boldsymbol{X}] \neq \mathbb{E}[Z \mid \boldsymbol{X}]$, then β_{IV} necessarily incorporates negatively weighted treatment effects for some groups and therefore fails even a minimal notion of causality.

Definition RC: Rich Covariates

Key requirement for LATE: For the LATE interpretation to hold, it is necessary that $L[Z \mid \boldsymbol{X}] = \mathbb{E}[Z \mid \boldsymbol{X}]$, a condition called *rich covariates*.

Definition (RC)

Let

$$L[Z \mid \boldsymbol{X} = x] \;\equiv\; \mathbb{E}[Z\boldsymbol{X}']\,\mathbb{E}[\boldsymbol{X}\boldsymbol{X}']^{-1}x$$

denote the population fitted value at $\boldsymbol{X} = x$ from regressing Z onto $\boldsymbol{X}$.

An IV specification is said to have *rich covariates* if

$$\mathbb{E}[Z \mid \boldsymbol{X} = x] = L[Z \mid \boldsymbol{X} = x] \qquad \text{for every } x \in \mathcal{X}.$$

That is, the conditional expectation of Z given $\boldsymbol{X}$ is exactly linear in $\boldsymbol{X}$.

Failure of rich covariates: β_{IV} reflects not only complier effects, but also *negatively weighted* always-taker effects.

Proposition 2: Alternative Decompositions of β_{IV}

Assumption: $\mathbb{E}[Y(t) \mid \boldsymbol{X}] = \eta_t^\top \boldsymbol{X}$, for $t \in \{0, 1\}$.

Never-taker conditional average treatment effect:

$$\Delta(\mathsf{NT}, x) \equiv \mathbb{E}[Y(1) - Y(0) \mid G = \mathsf{NT}, \boldsymbol{X} = x].$$

Result (for any $\varepsilon \in \mathbb{R}$)

$$\begin{aligned}\beta_{IV} = &\ \mathbb{E}[\omega_\varepsilon(\mathsf{CP}, \boldsymbol{X})\Delta(\mathsf{CP}, \boldsymbol{X})] \\ &+ \mathbb{E}[\omega_\varepsilon(\mathsf{AT}, \boldsymbol{X})\Delta(\mathsf{AT}, \boldsymbol{X})] \\ &+ \mathbb{E}[\omega_\varepsilon(\mathsf{NT}, \boldsymbol{X})\Delta(\mathsf{NT}, \boldsymbol{X})]\,.\end{aligned}$$

Key point: β_{IV} admits a continuum of valid decompositions indexed by ε.

Proposition 2: Weights and Interpretation

Weights (for any $\varepsilon \in \mathbb{R}$)

$$\omega_\varepsilon(\mathsf{CP}, \boldsymbol{X}) \equiv \Big(\varepsilon\, \mathbb{E}[\tilde{Z} \mid \boldsymbol{X}] + L[Z \mid \boldsymbol{X}]\big(1 - \mathbb{E}[Z \mid \boldsymbol{X}]\big)\Big) \times \Pr(G = \mathsf{CP} \mid \boldsymbol{X})\mathbb{E}[\tilde{Z}T]^{-1},$$

$$\omega_\varepsilon(\mathsf{AT}, \boldsymbol{X}) \equiv \varepsilon\, \mathbb{E}[\tilde{Z} \mid \boldsymbol{X}] \Pr(G = \mathsf{AT} \mid \boldsymbol{X})\mathbb{E}[\tilde{Z}T]^{-1},$$

$$\omega_\varepsilon(\mathsf{NT}, \boldsymbol{X}) \equiv (\varepsilon - 1)\, \mathbb{E}[\tilde{Z} \mid \boldsymbol{X}] \Pr(G = \mathsf{NT} \mid \boldsymbol{X})\mathbb{E}[\tilde{Z}T]^{-1}.$$

Proposition 1 corresponds to $\varepsilon = 1$.

Key implication: Unless rich covariates hold, every choice of ε leads to an interpretation involving

- always-takers, or never-takers, or both,
- negative weights for some values of $\boldsymbol{X}$,
- potentially negative weights even for compliers.

Proof for Proposition 1

Step 1: Decomposing the Estimator

The Decomposition: Using the residual $\tilde{Z} = Z - L[Z \mid \boldsymbol{X}]$, we split the numerator $\mathbb{E}[Y\tilde{Z}]$ into two parts:

$$\mathbb{E}[Y\tilde{Z}] = \underbrace{\mathbb{E}[\mathrm{Cov}(Y, Z \mid \boldsymbol{X})]}_{\text{Standard IV Term (A)}} + \underbrace{\mathbb{E}\big[\mathbb{E}[Y \mid \boldsymbol{X}]\mathbb{E}[\tilde{Z} \mid \boldsymbol{X}]\big]}_{\text{Specification Error (B)}}. \tag{1}$$

Analysis of Term A (Standard IV): Conditional on $\boldsymbol{X}$, this term captures only compliers (Imbens & Angrist, 1994). For a binary instrument:

$$\mathrm{Cov}(Y, Z \mid \boldsymbol{X}) = \Delta(\mathsf{CP}, \boldsymbol{X})\,\Pr(G = \mathsf{CP} \mid \boldsymbol{X})\,\underbrace{\mathbb{E}[Z \mid \boldsymbol{X}](1 - \mathbb{E}[Z \mid \boldsymbol{X}])}_{\mathrm{Var}(Z|\boldsymbol{X})}.$$

Intuition

The first term is the "good" part: it represents the causal effect of compliers, weighted by the variance of the instrument. If $\mathbb{E}[\tilde{Z} \mid \boldsymbol{X}] = 0$ (i.e., "rich covariates"), the second term vanishes, and we perfectly recover a weighted LATE.

Step 2: Expanding $\mathbb{E}[Y \mid \boldsymbol{X}]$

The Problem: The error term in (1) depends on outcome levels $\mathbb{E}[Y \mid \boldsymbol{X}]$. We decompose this by group:

$$\mathbb{E}[Y \mid \boldsymbol{X}] = \mathbb{E}[Y(0) \mid \mathsf{NT}] \Pr(\mathsf{NT} \mid \boldsymbol{X}) + \mathbb{E}[Y(1) \mid \mathsf{AT}] \Pr(\mathsf{AT} \mid \boldsymbol{X}) + \dots$$

The Centering Trick ($\varepsilon = 1$): To isolate the bias, we "center" the expression around the untreated baseline $\mathbb{E}[Y(0) \mid \boldsymbol{X}]$. Assuming linearity $\mathbb{E}[Y(0) \mid \boldsymbol{X}] = \eta_0^\top \boldsymbol{X}$, the expansion simplifies to:

$$\mathbb{E}[Y \mid \boldsymbol{X}] = \underbrace{\Delta(\mathsf{CP}, \boldsymbol{X}) \Pr(\mathsf{CP} \mid \boldsymbol{X}) \mathbb{E}[Z \mid \boldsymbol{X}]}_{\text{Complier Contribution}} + \underbrace{\Delta(\mathsf{AT}, \boldsymbol{X}) \Pr(\mathsf{AT} \mid \boldsymbol{X})}_{\text{Always-Taker Contribution}} + \underbrace{\eta_0^\top \boldsymbol{X}}_{\text{Linear Baseline}} . \tag{2}$$

Intuition

By centering around the $Y(0)$ trend, we isolate the **excess** outcomes. Always-Takers remain because their outcome $Y(1)$ differs from the baseline $Y(0)$. Never-Takers disappear because their outcome $Y(0)$ is fully absorbed into the linear baseline $\eta_0^\top \boldsymbol{X}$.

Step 3: Orthogonality and Final Result

Exploiting Orthogonality: Since $\tilde{Z}$ is a linear residual, it is orthogonal to the linear baseline $\eta_0^\top \boldsymbol{X}$:

$$\mathbb{E}[(\eta_0^\top \boldsymbol{X})\tilde{Z}] = 0.$$

Substituting (2) back into (1), the linear trend vanishes. The bias term $\mathbb{E}[\tilde{Z} \mid \boldsymbol{X}]$ interacts only with the remaining groups:

$$\mathbb{E}[Y\tilde{Z}] = \text{Term A} + \mathbb{E}\Big[\big(\text{Compliers} + \text{Always-Takers}\big)\mathbb{E}[\tilde{Z} \mid \boldsymbol{X}]\Big].$$

Dividing by the denominator $\mathbb{E}[T\tilde{Z}]$ yields Proposition 1:

$$\beta_{IV} = \mathbb{E}[\omega(\text{CP}, \boldsymbol{X})\Delta(\text{CP}, \boldsymbol{X})] + \mathbb{E}[\omega(\text{AT}, \boldsymbol{X})\Delta(\text{AT}, \boldsymbol{X})].$$

Conclusion

Always-Takers receive non-zero weight specifically because the linear control $\eta_0^\top \boldsymbol{X}$ fails to perfectly orthogonalize $\tilde{Z}$ against the outcome levels (i.e., $\mathbb{E}[\tilde{Z} \mid \boldsymbol{X}] \neq 0$).

Q.E.D.

Weaker Notion of Causality

Setup: Group Treatment Responses

We work in a nonparametric IV framework with potential outcomes and define the *group treatment responses (GTRs)* as

$$\mu_j(g, x) \equiv \mathbb{E}[Y(t_j) \mid G = g,\ \boldsymbol{X} = x]\,,$$

the expected potential outcomes across treatment levels, compliance groups, and covariates.

Collect all GTRs as

$$\mu = \{\mu_j(g, x) : j = 0, 1, \ldots, J,\ g \in \mathcal{G},\ x \in \mathcal{X}\},$$

which takes values in $\mathbb{R}^{d_\mu}$.

Let β be any estimand whose value depends on μ.

Definition WC: Weakly Causal Estimand

Definition (WC).

An estimand β is *weakly causal* if, for any μ, both of the following hold:

$$\text{If } \mu_j(g,x) - \mu_{j-1}(g,x) \geq 0 \;\; \forall j \geq 1,\; g \in \mathcal{G},\; x \in \mathcal{X}, \quad \text{then } \beta \geq 0,$$

$$\text{If } \mu_j(g,x) - \mu_{j-1}(g,x) \leq 0 \;\; \forall j \geq 1,\; g \in \mathcal{G},\; x \in \mathcal{X}, \quad \text{then } \beta \leq 0. \tag{4}$$

Interpretation: If all underlying treatment effects have the same sign across all groups and covariates, then the estimand β must have that same sign.

Thus, a weakly causal estimand is not systematically misleading about the sign of the underlying treatment effects.

Theorem 1: Weak Causality of Linear IV

Assumption EXO (Exogeneity):

$$(G, \{Y(t)\}_{t \in \mathcal{T}}) \perp Z \mid \boldsymbol{X}.$$

Assumption MON (Monotonicity): For ordered instrument values $z_0 \leq z_1 \leq \cdots \leq z_K$,

$$\Pr\big(T(z_0) \leq T(z_1) \leq \cdots \leq T(z_K) \mid \boldsymbol{X} = x\big) = 1 \quad \forall x.$$

Theorem 1.

Suppose that Assumptions EXO and MON are satisfied. Then the linear IV estimand β_{IV} is *weakly causal* if and only if the IV specification has *rich covariates*.

Interpretation: Rich covariates are both *necessary* and *sufficient* for β_{IV} to preserve the sign of underlying treatment effects.

When Do Rich Covariates Hold?

Two important special cases (Kolesár, 2013):

1. Saturated covariates: If $\boldsymbol{X}$ is a saturated specification (a full set of indicators for mutually exclusive and exhaustive events), then the IV specification has rich covariates.

2. Mean independence: If Z is mean independent of $\boldsymbol{X}$, so that

$$\mathbb{E}[Z \mid \boldsymbol{X} = x] = \mathbb{E}[Z] \quad \text{for all } x,$$

then

$$L[Z \mid \boldsymbol{X} = x] \;\equiv\; \mathbb{E}[Z\boldsymbol{X}']\,\mathbb{E}[\boldsymbol{X}\boldsymbol{X}']^{-1}x \;=\; \mathbb{E}[Z]\,\mathbb{E}[\mathbf{1}\boldsymbol{X}']\,\mathbb{E}[\boldsymbol{X}\boldsymbol{X}']^{-1}x \;=\; \mathbb{E}[Z],$$

since $\boldsymbol{X}$ contains a constant.

Outside these cases: The assumption that an IV specification has rich covariates is an implicit *parametric restriction* and must be explicitly justified.

Implications for Applied IV Work

Main lesson for applications

- Linear IV with covariates should *not* be interpreted as LATE by default.
- A causal interpretation requires **rich covariates**:

$$\mathbb{E}[Z \mid X] = L[Z \mid X].$$

- When covariates are not rich, TSLS may:
  - place negative weights on treatment effects, or
  - depend on levels of potential outcomes.

Practical guidance from the paper

- Saturated specifications ("saturate and weight") satisfy rich covariates.
- Designs where Z is mean independent of X also satisfy rich covariates.
- Outside these cases, rich covariates is a parametric assumption that must be justified.

Alternatives and Interpretation

If rich covariates are implausible

- Avoid interpreting TSLS as LATE or even as a causal average.
- Explicitly interpret TSLS as a weighted average that may include:
  - negative weights, or
  - dependence on outcome levels.

Alternative approaches discussed in the paper

- Nonparametric IV methods that condition directly on X.
- Fully saturated covariate specifications.
- Designs with randomized or mean-independent instruments.

Final takeaway

- Rich covariates are *necessary and sufficient* for TSLS to be weakly causal.
- Without them, TSLS may fail even minimal causal sign requirements.

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